

Models comparison

“Whatever you can do or dream you can, begin it; boldness has genius, power, and magic in it.”

W. H. Murray

Common spatial extent/locations

Temporal span: 2015-2020

Spatial resolution: 5.5 km

CERRA Clim



XXX models: sPop, ArboCartoR, dynamAedes, VECTRI

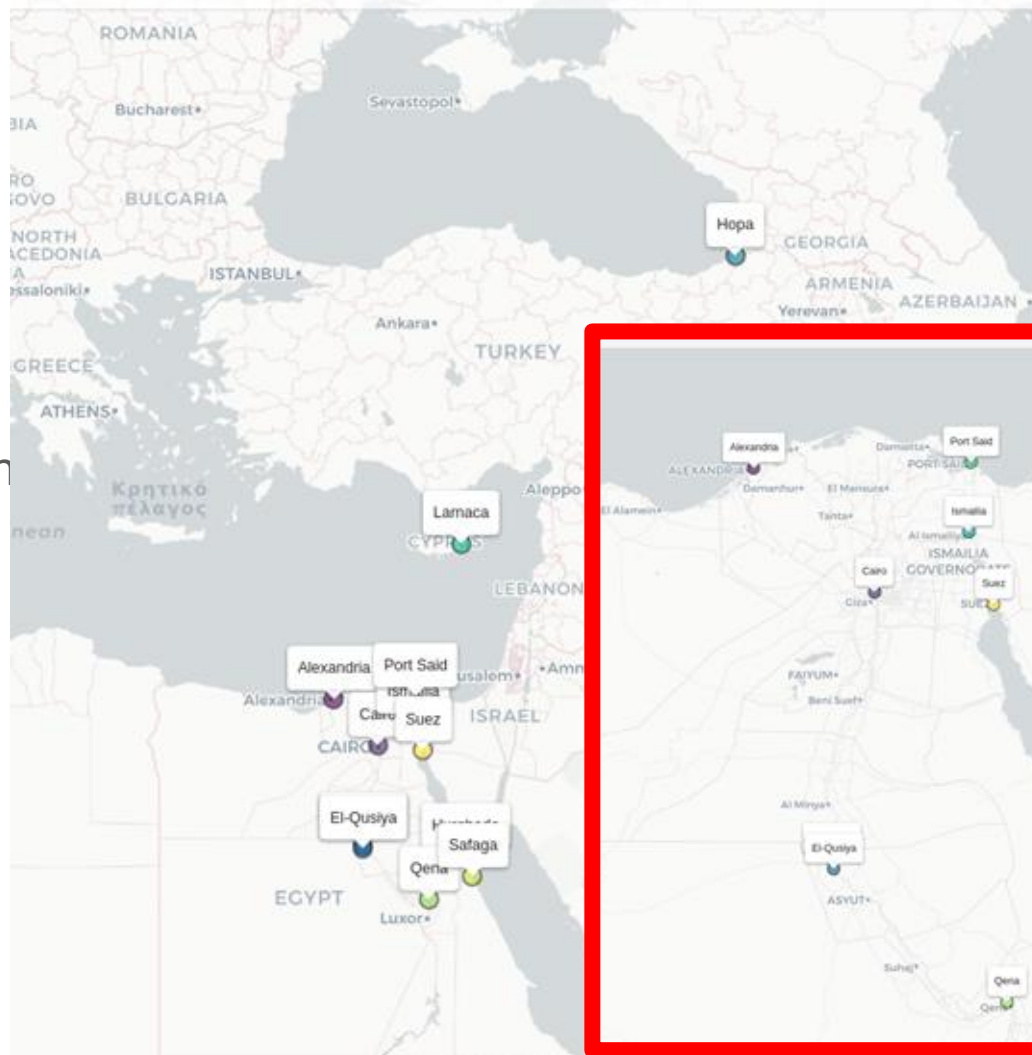
+ Benedict's Model + Sadie's Model + Jesus' model + Mina's model

other models I might have missed?

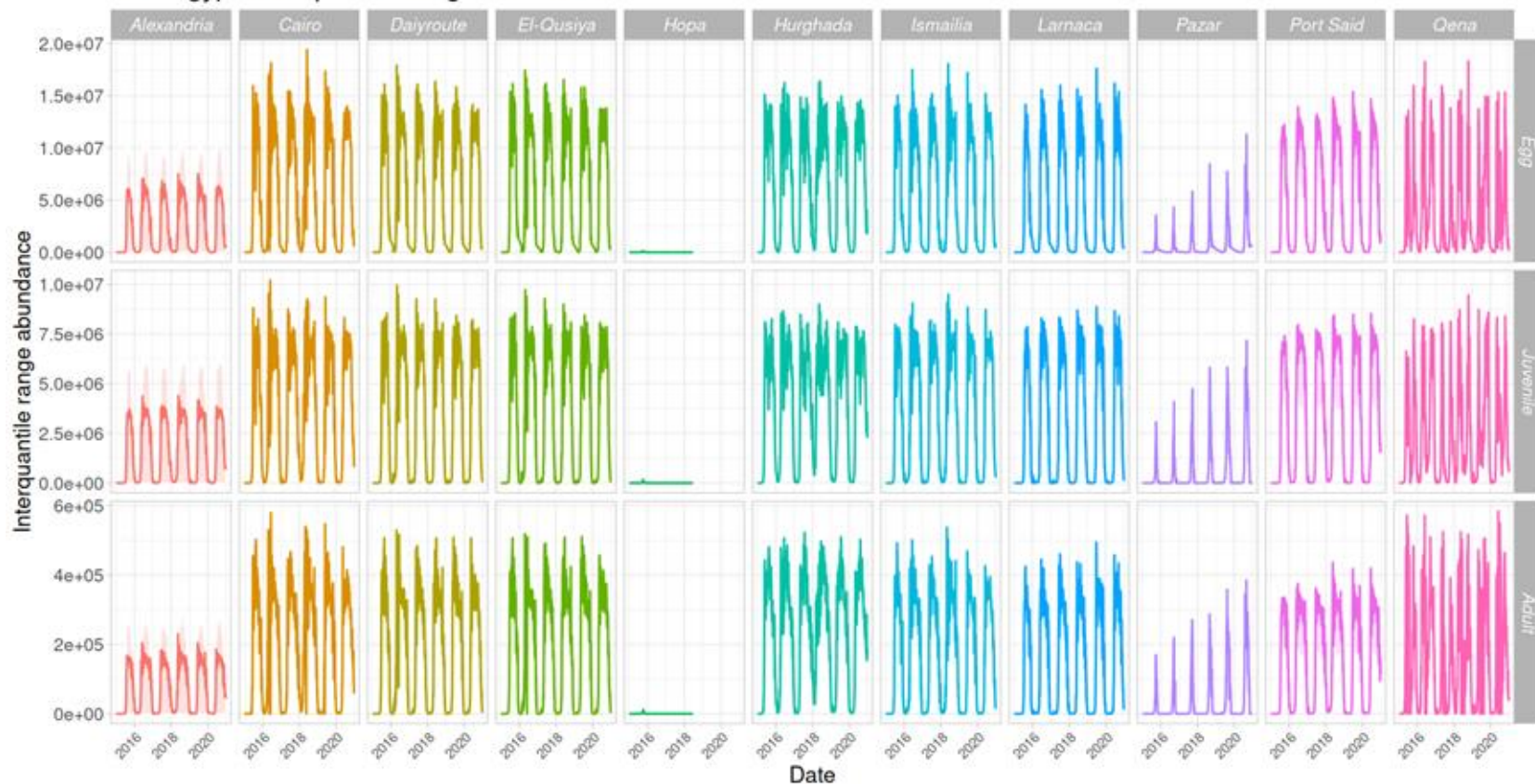
We focus on predicted adults

dynamAedes

- stochastic model
- E + J + A stages
- different geographical scales
- originally designed for invasion scenarios
- dedicated R library



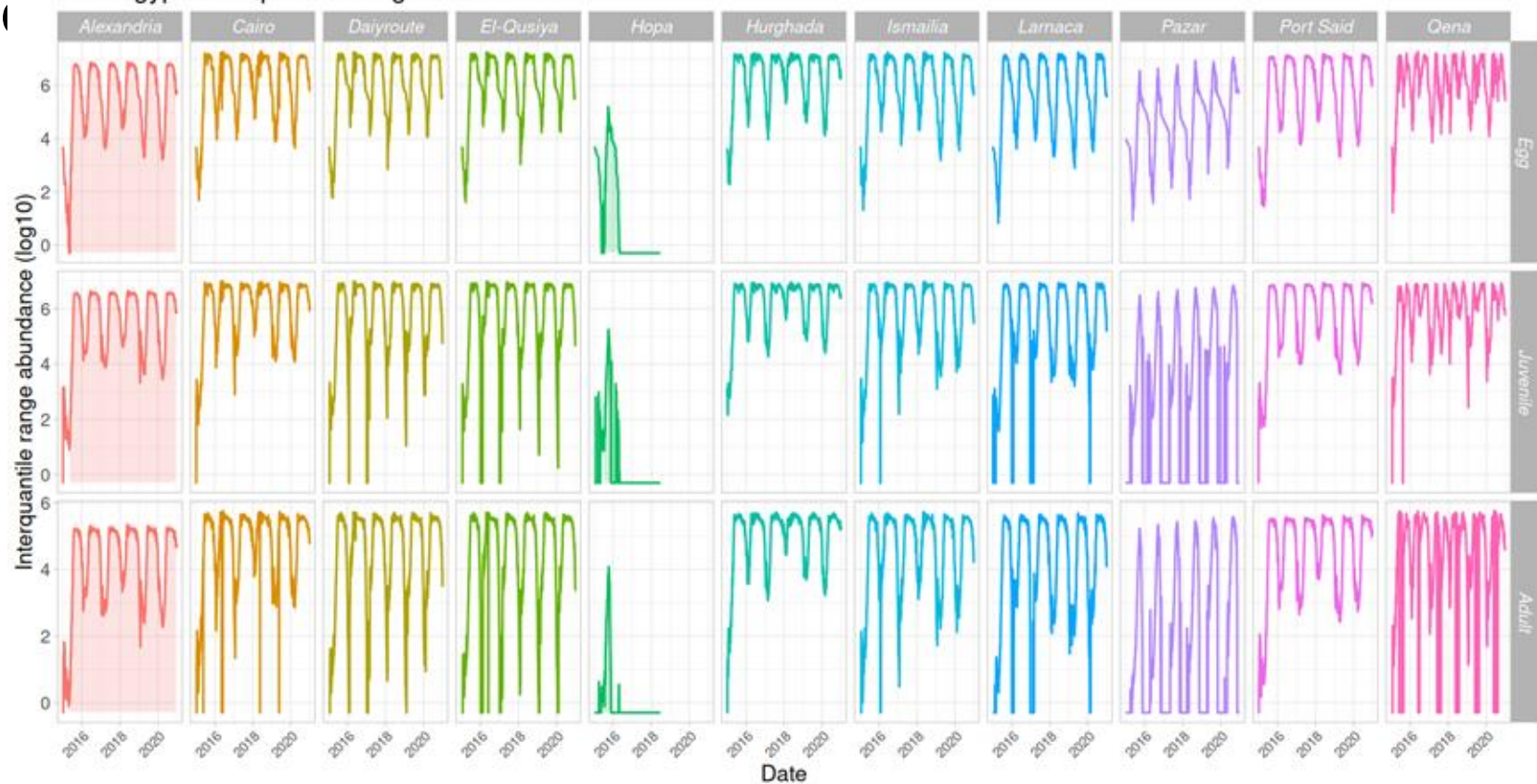
Ae. aegypti Interquantile range abundance



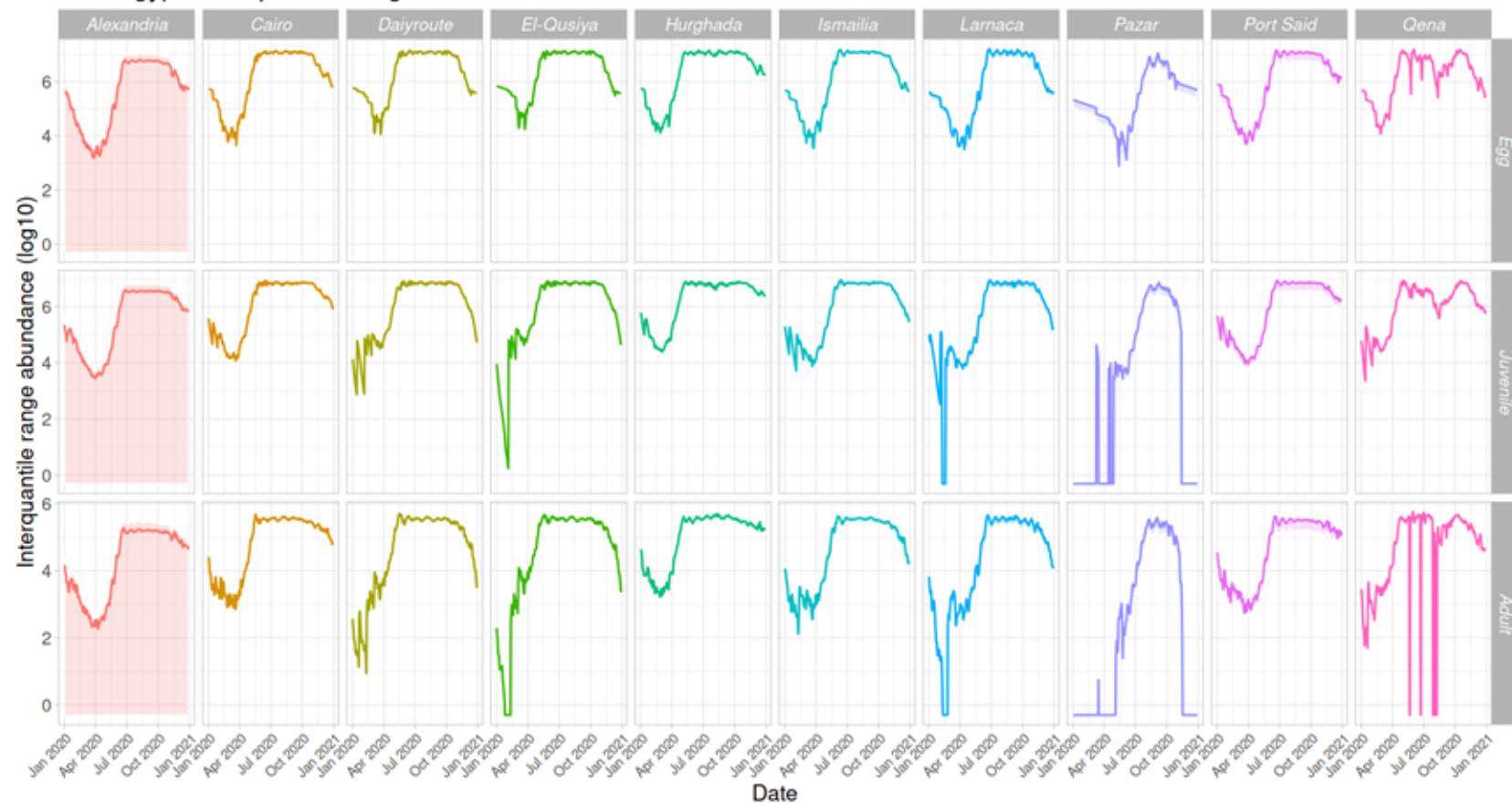
Locations

- Alexandria
- Cairo
- Daiyroute
- El-Qusiya
- Hopa
- Hurghada
- Ismailia
- Larnaca
- Pazar
- Port Said
- Qena

Ae. aegypti Interquartile range abundance



Ae. aegypti Interquantile range abundance

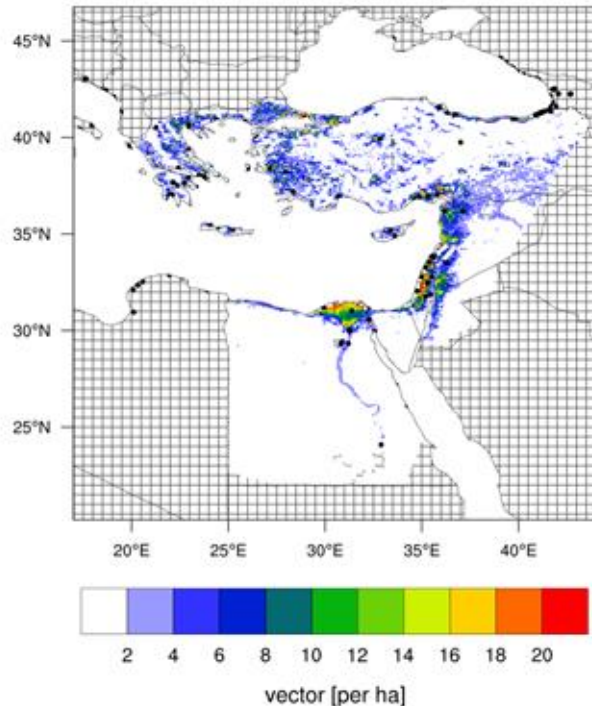


VECTRI

- Dynamical disease model originally developed for malaria & *An. gambiae* ([Tompkins et al. 2013](#))
- Eggs + Larval + Adult stages depend on temperature/rainfall
- Gridded simulations (time,lat,lon). Inputs: daily rainfall and temperature data and population density.
- Code written in Fortran 90, netcdf or grib climate data files as inputs
- *Aedes aegypti* model based on *Ae. albopictus* model ([Garrido Zornoza M., et al. 2024](#)) with larval and adult mortality schemes based on [Caldwell. et al. 2021](#)
- Early prototype, spatial density tends to be very low for adults. Validation in progress for Europe and Colombia

VECTRI: preliminary results

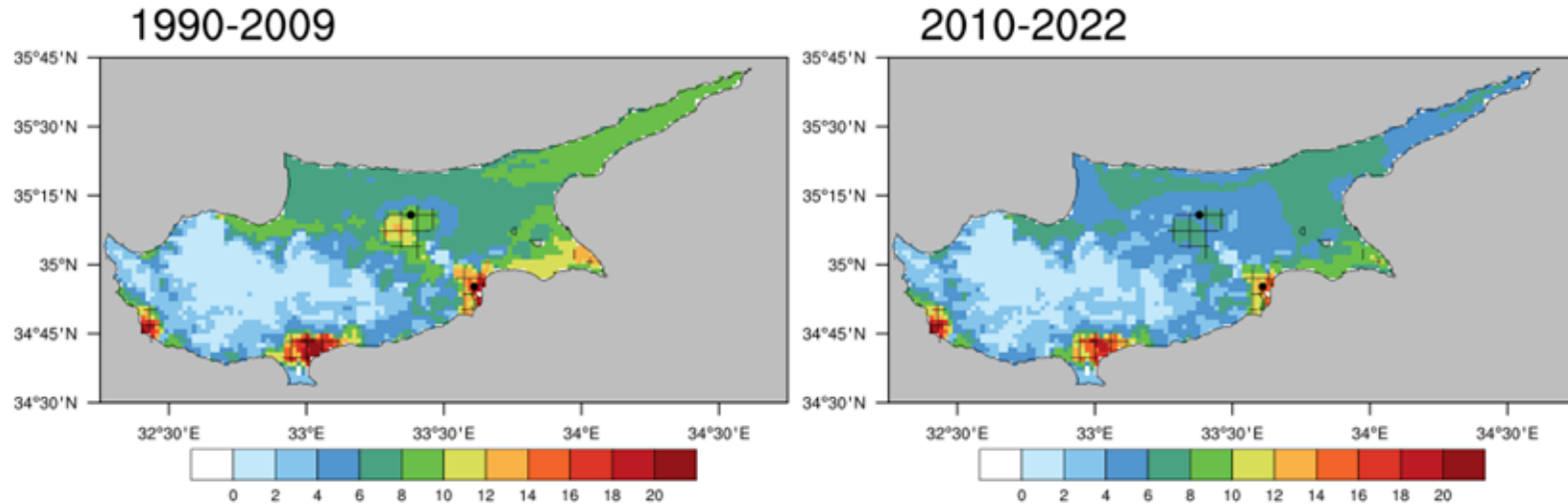
Vectri CERRA 2015-2020



Simulated adult Ae. aegypti density (per ha) by the VECTRI model for the period 2015-2020 driven by CERRA climate data. Black dots depict historical occurrences based on [Wint et al., 2022](#).

Hotspots are simulated in the vicinity of the Nile river, and coastal areas of the south-eastern Mediterranean region. Good match between simulations and observed historical occurrence of *Ae. aegypti* over the region. validation scores to do (AUC etc).

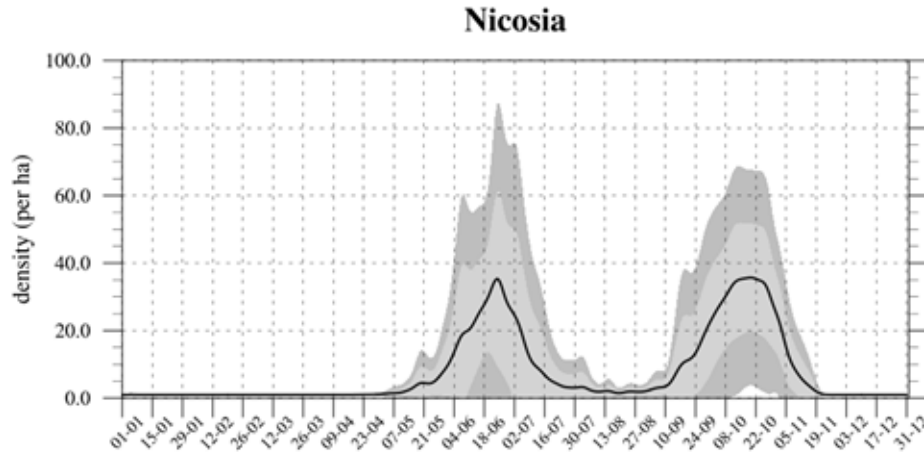
VECTRI: preliminary results, 1 km simulations (EMO)



Simulated adult Ae. aegypti density (per ha) by the VECTRI model for the periods 1990-2009 and 2010-22. Hatching denotes areas > 250 person/km² (GPWv4 dataset). The model was driven by the EMO rainfall and temperature data available at <https://doi.org/10.2905/0BD84BE4-CEC8-4180-97A6-8B3ADAAAC4D26>

Hotspots simulated over densely populated regions of Cyprus: Paphos, Limassol, Larnaca, Nicosia and Ayia Napa. Moderate risk simulated over the northern coasts. Decrease in simulated abundance during the past 10-15y, excepting over Paphos.

VECTRI: preliminary results



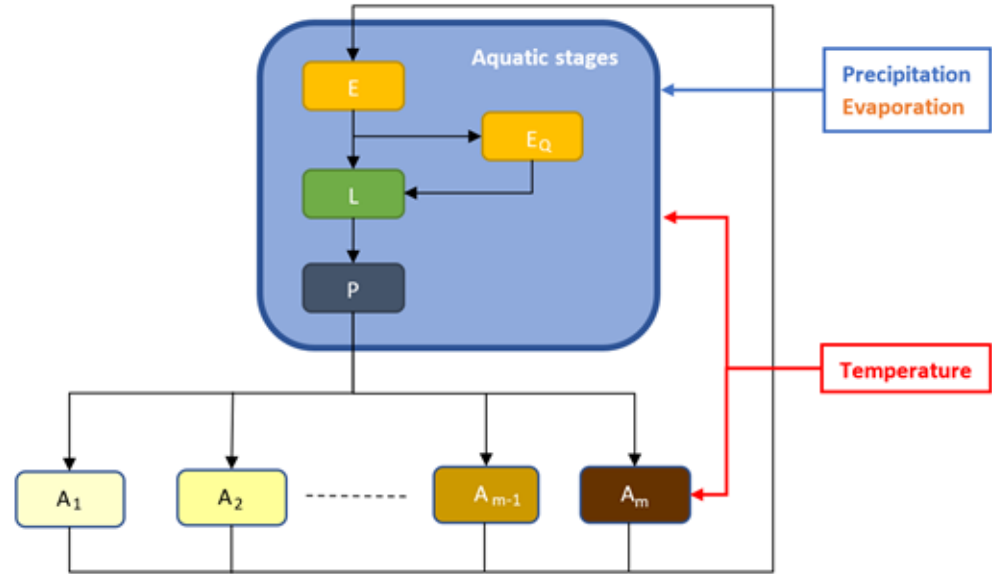
Simulated vector activity in Nicosia is bimodal and shows a peak in June-July and one peak in Oct-Nov.

Warm conditions in summer tend to increase simulated mortality rates in Nicosia.

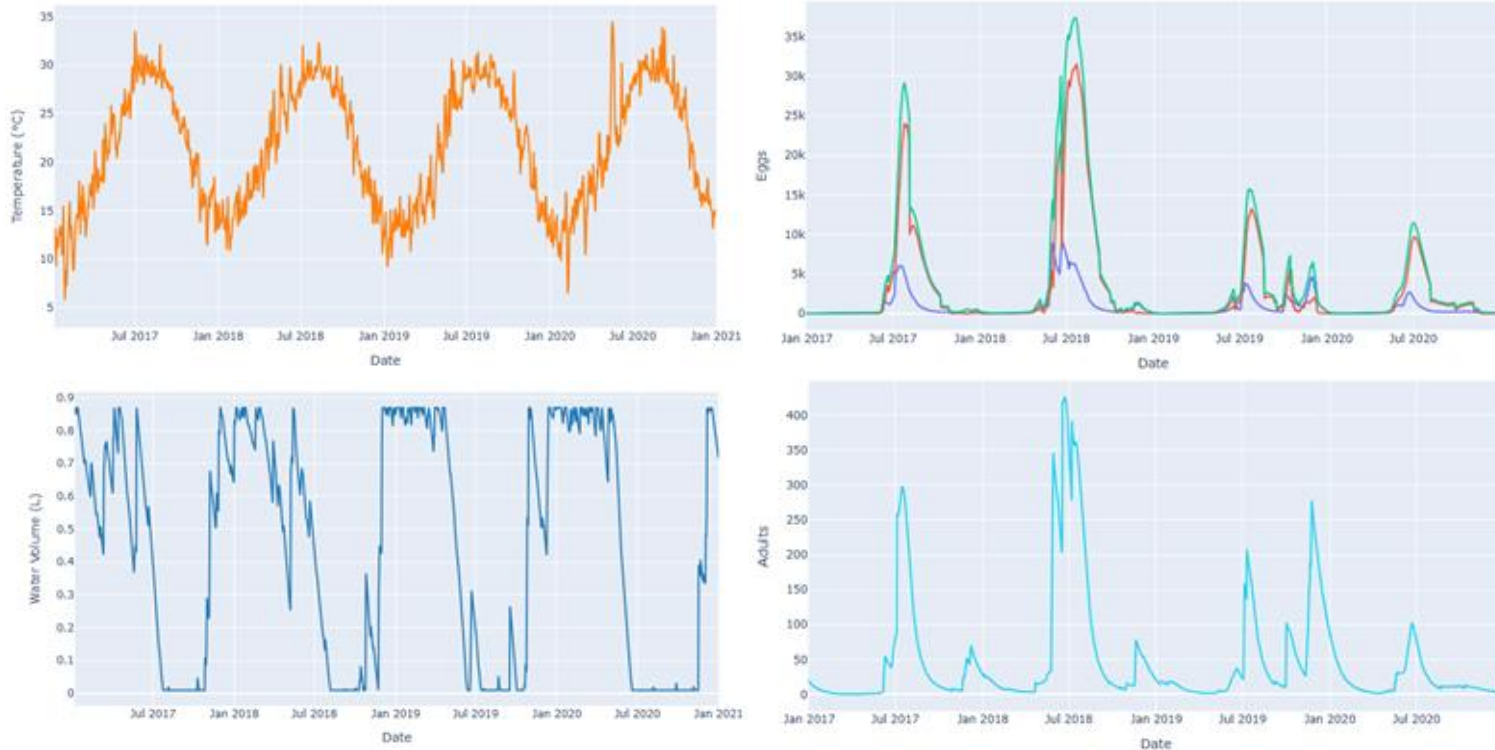
Mean seasonal cycle of simulated adult density (Ae. aegypti) in Nicosia (CERRA simulation). The mean for 2015-2020 is depicted by the black solid line. The light-gray and gray envelope respectively show one and two standard deviations around the mean calculated for all years.

Delay-differential model for *Ae. aegypti*

- Uses an environmentally variable delay-differential equation
- Models egg hatching in a naturally filled larval habitat
- Has size-dependent fecundity in the adult stage
- Using CERRA data for temperature and precipitation (Estimating evaporation from relative humidity/temperature)

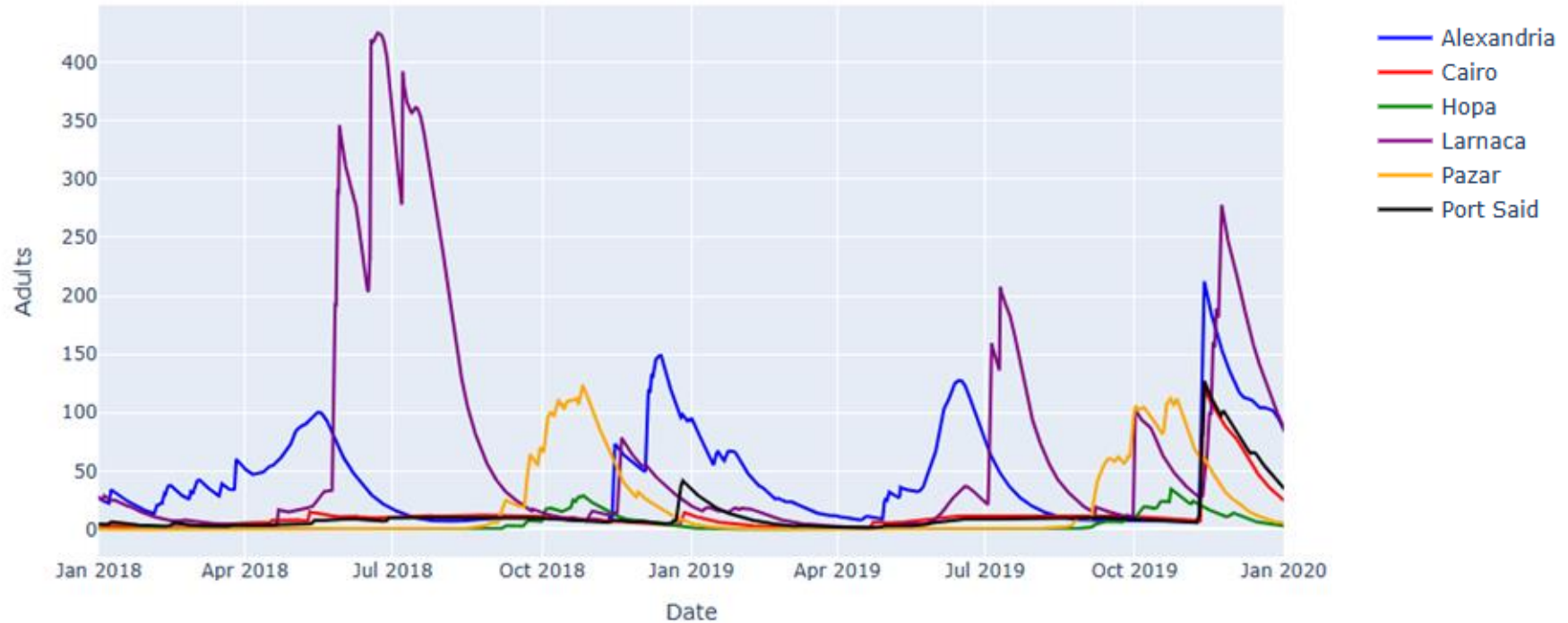


Preliminary results Larnaca - DDE for *Ae. aegypti*



Model predicts both dependence on the water dynamics and temperature with peaks at the start and end of the container being filled (usually July and November).

Preliminary comparisons between cities - DDE for *Ae. aegypti*



Most the cities did not having the container filling so other water sources

MOBVEC model: (Proof of concept / Early development)

Goal: Model mosquito movement in the early stages of a VBD outbreak to optimize trap placement.

$$\begin{aligned}\partial_t E(x,y,t) &= \alpha|_{bs} A - (d_E \cdot h + \delta_E) E \\ \partial_t L(x,y,t) &= d_E \cdot h E - d_L L - \delta_L L - L^2 / K \\ \partial_t A(x,y,t) &= d_L L - \delta_A A + \nabla(D \cdot \nabla A) - \gamma|_{tr} A \\ \partial_t Tr(x,y,t) &= \gamma|_{tr} A\end{aligned}$$

$\alpha, d_E, d_L, \delta_E, \delta_L, \delta_A$: Thermal biology, Mordecai et al. (2017), Pardo-Araujo et al. (2024), own fits.

h, K : Function of R and H, Metelmann et al. (2019), Erguler et al. (2016)

D : Land cover, ESA

Time scale: Weeks.

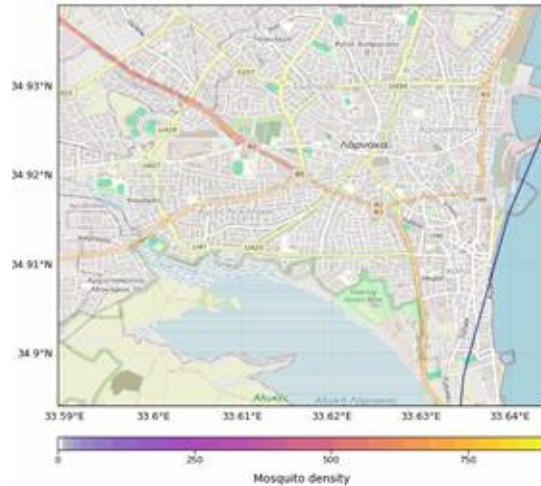
Spatial scale: Few km².

$|_{bs}$: Reports $|_{tr}$: Optimization

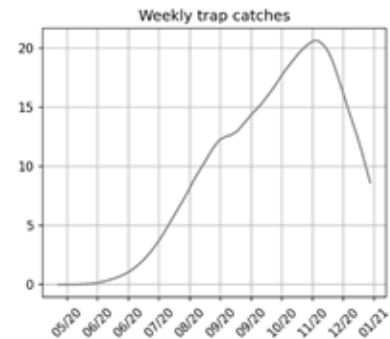
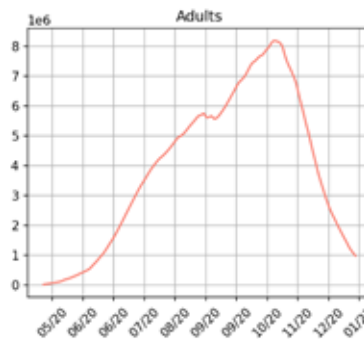
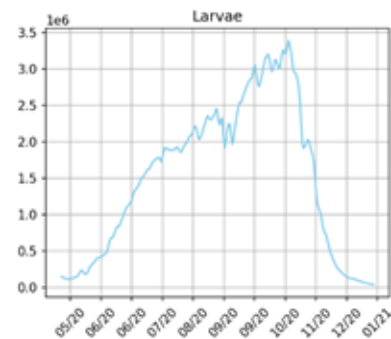
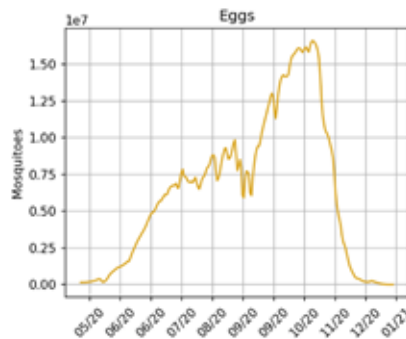
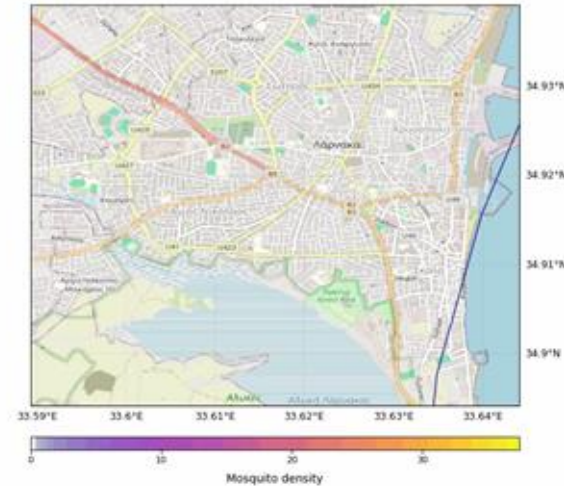
MOBVEC model: (Proof of concept / Early development)

01/05/2020

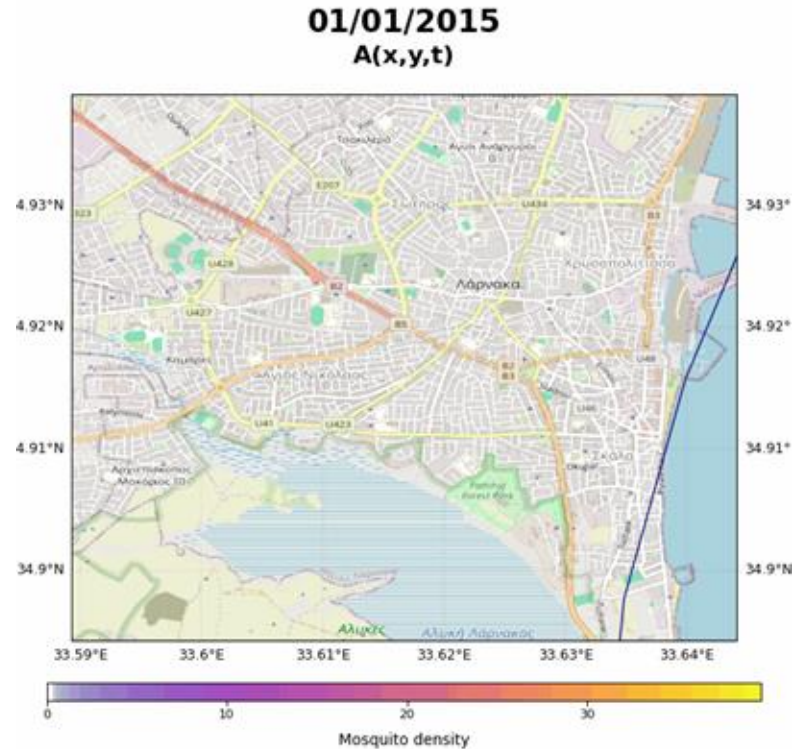
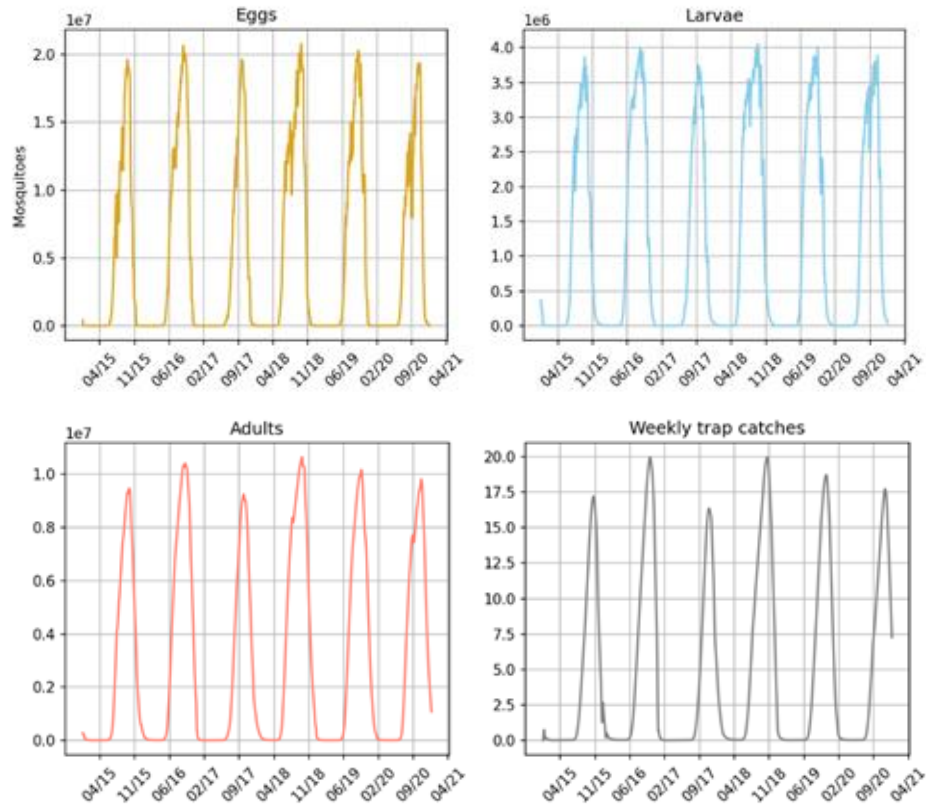
$E(x,y,t)$



$A(x,y,t)$

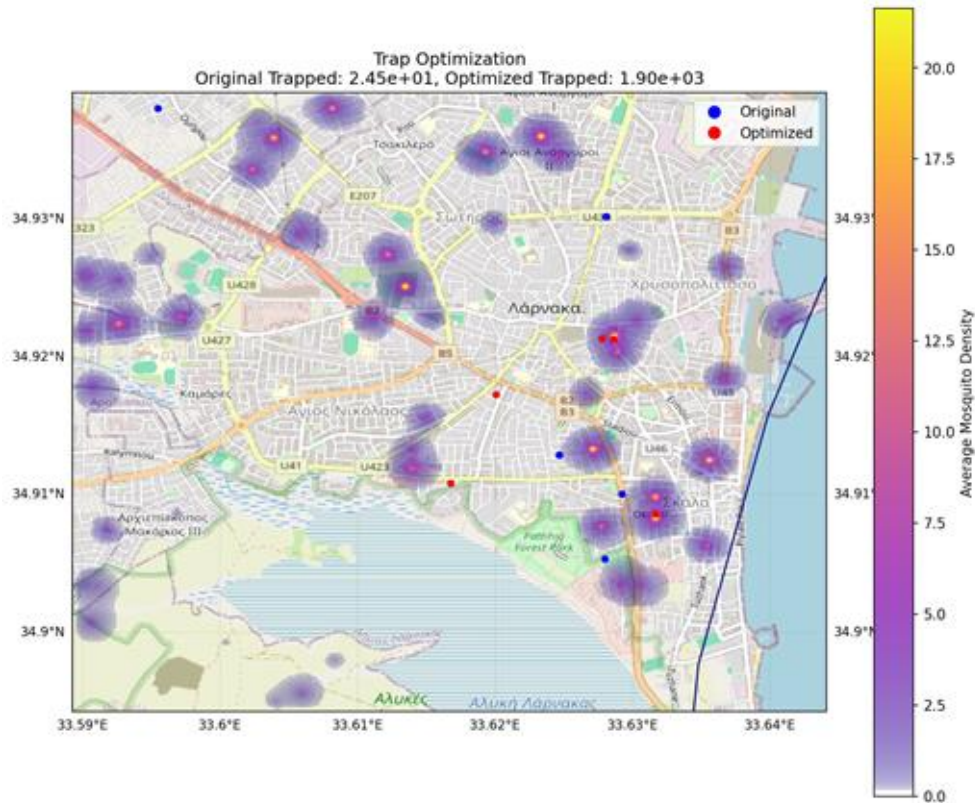
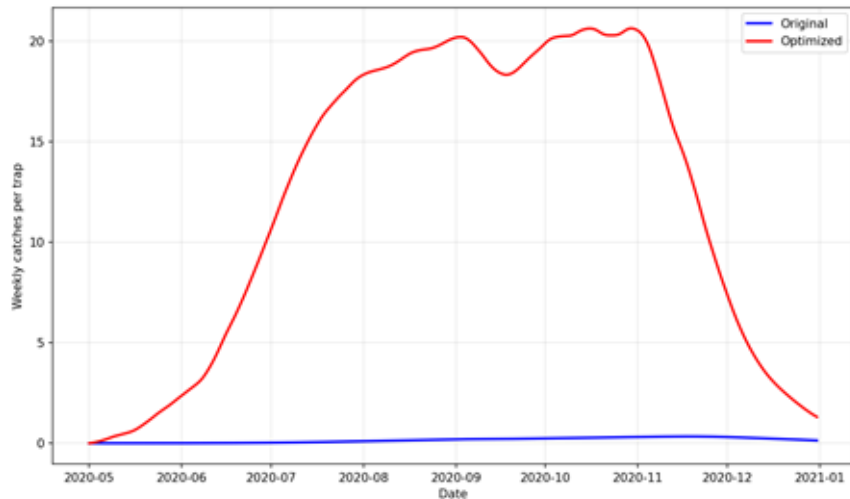


MOBVEC model: (Proof of concept / Early development)

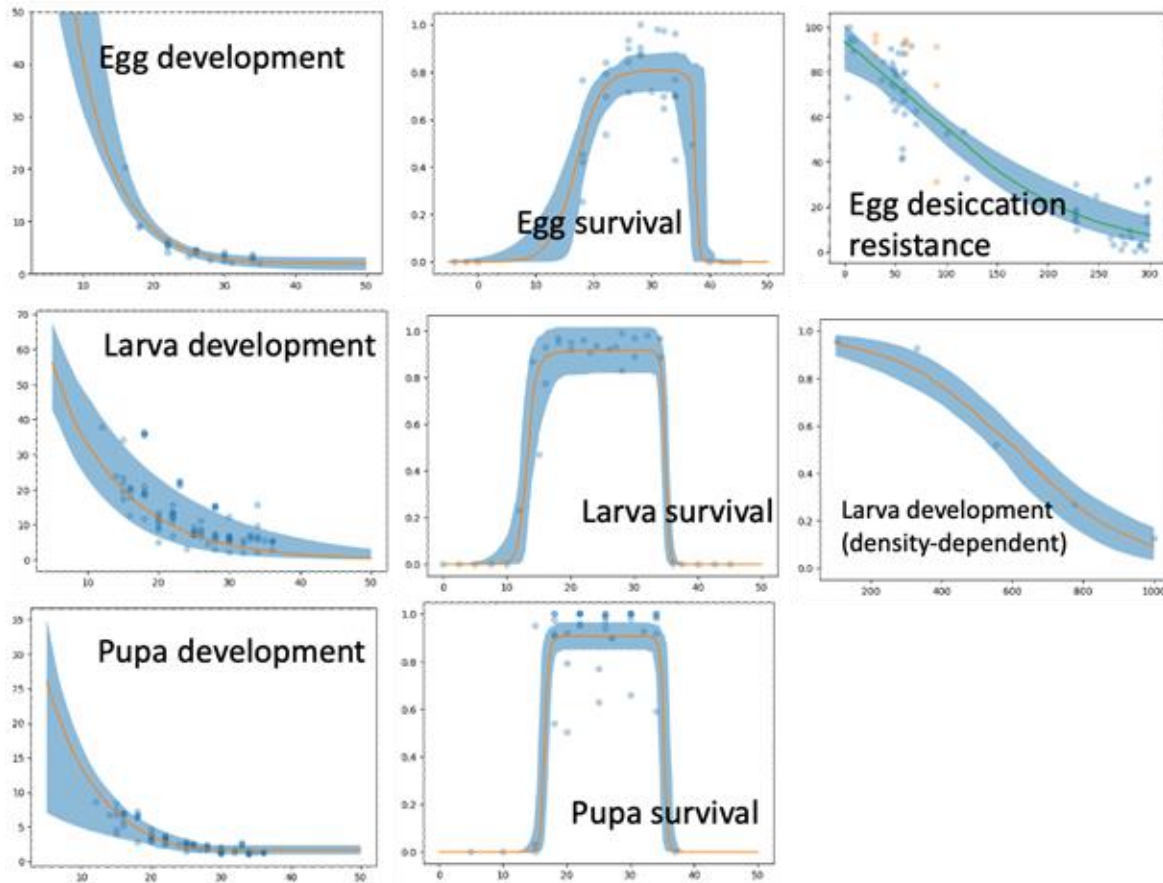


MOBVEC model: (Proof of concept / Early development)

Trap placement optimization



sPop2 – aegypti model



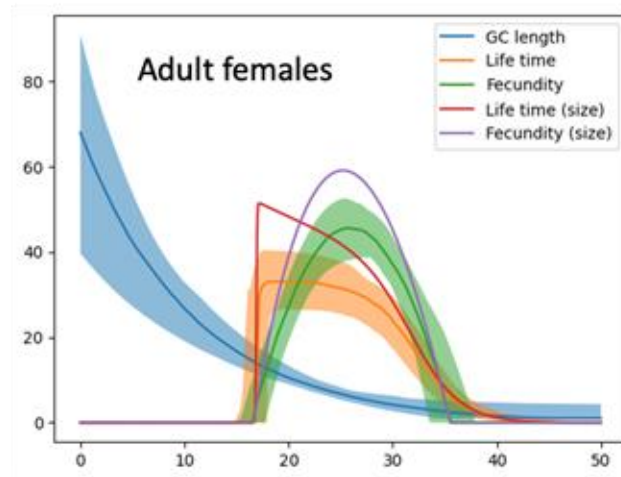
VecTraits



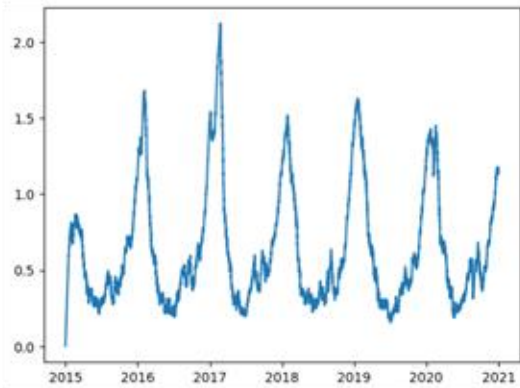
Kivanc Sevim



Bulent Alten



sPop2 – aegypti model

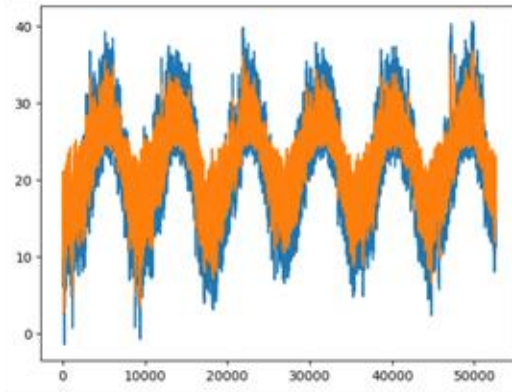


Breeding site volume as a function of

- Precipitation
- Human population density
- Evaporation

To account for **adaptive behaviour**

- Extracted optimum temperature within k standard deviations of the observed temperature, where k grows smoothly with distance from T_{opt}

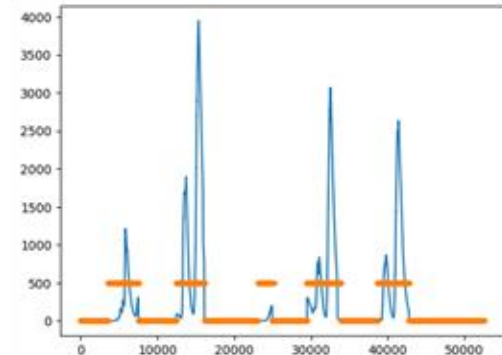


Optimised

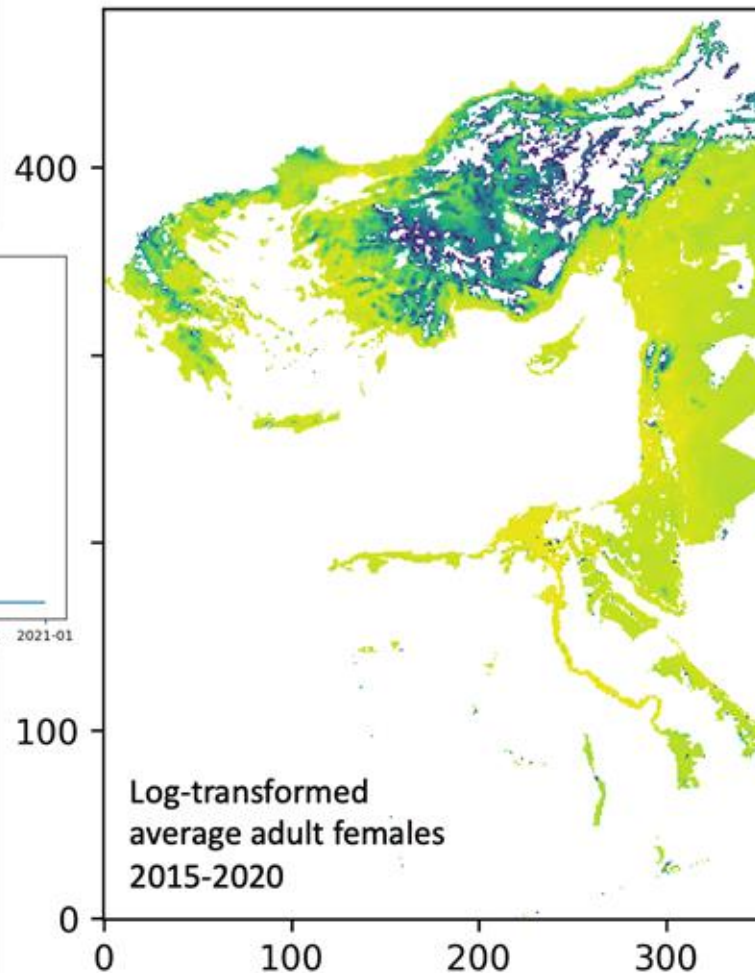
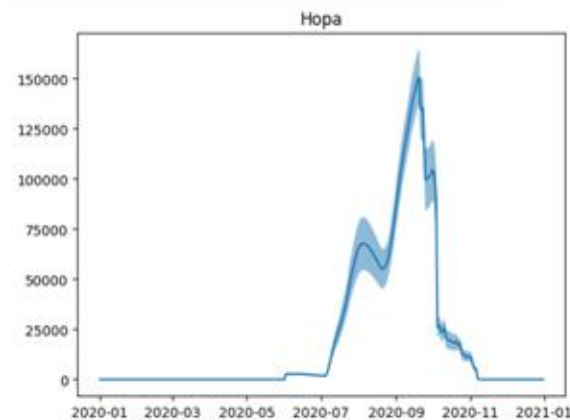
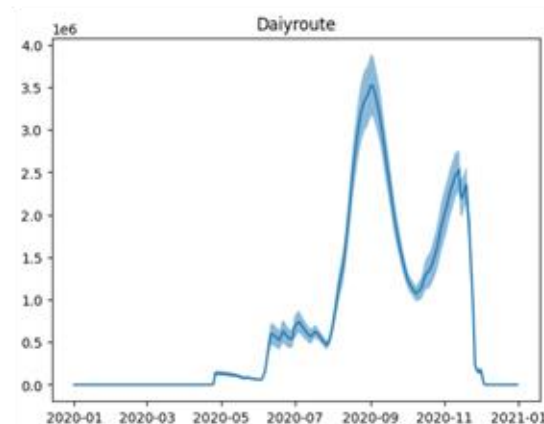
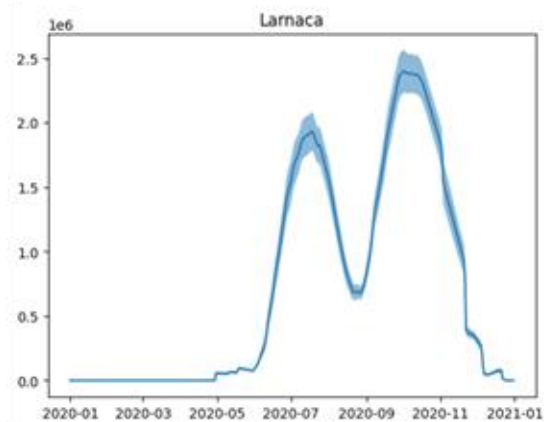
- Breeding site control coefficients for
 - Precipitation
 - Human population density
 - Evaporation
- Egg hatching queues
 - Time for activation
 - Minimum breeding site volume
 - Minimum temperature

For

- The number of days with adults female presence in 3 locations with known presence
 - Larnaca, Cyprus
 - Hopa, Türkiye
 - Daiyroute, Egypt



sPop2 – aegypti model



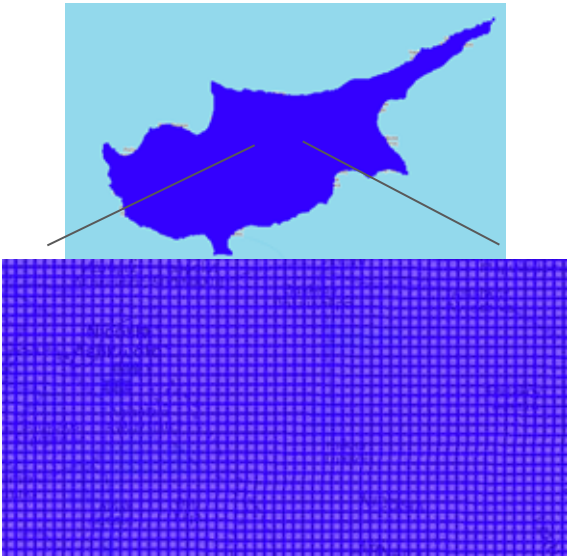
Alternative model 3

Objective:
Improving the classification (Aedes),
expanding the sample size, refining the data
resolution, and including the time factor

- Data:
- Positive trap locations: 17 points (Cyprus)
 - Negative trap locations: 51 points (Cyprus)
 - 5 time series data, 9 static data

Date considered:

- 2023 August and July 2024

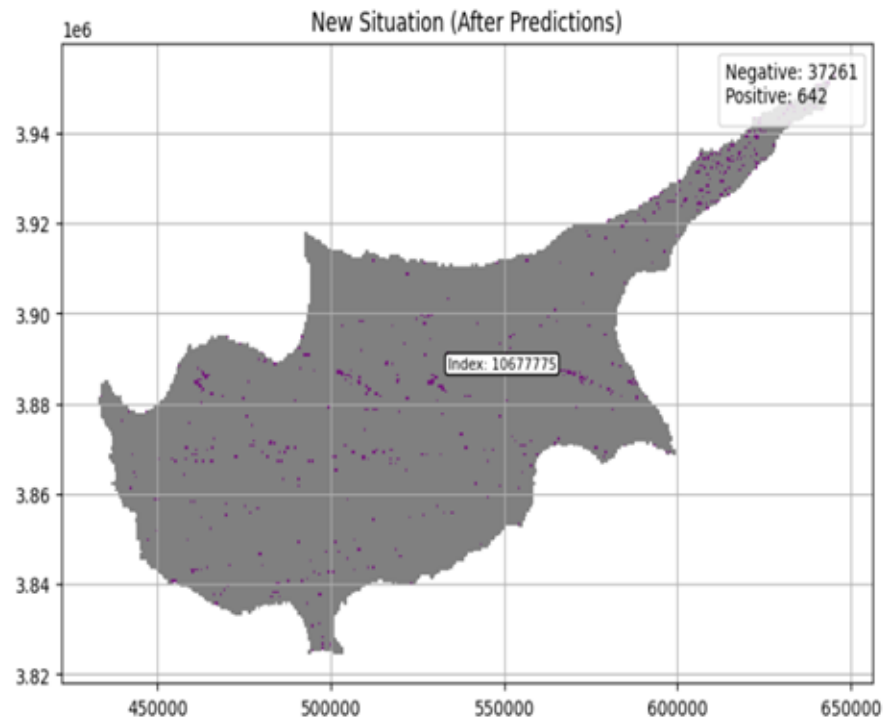
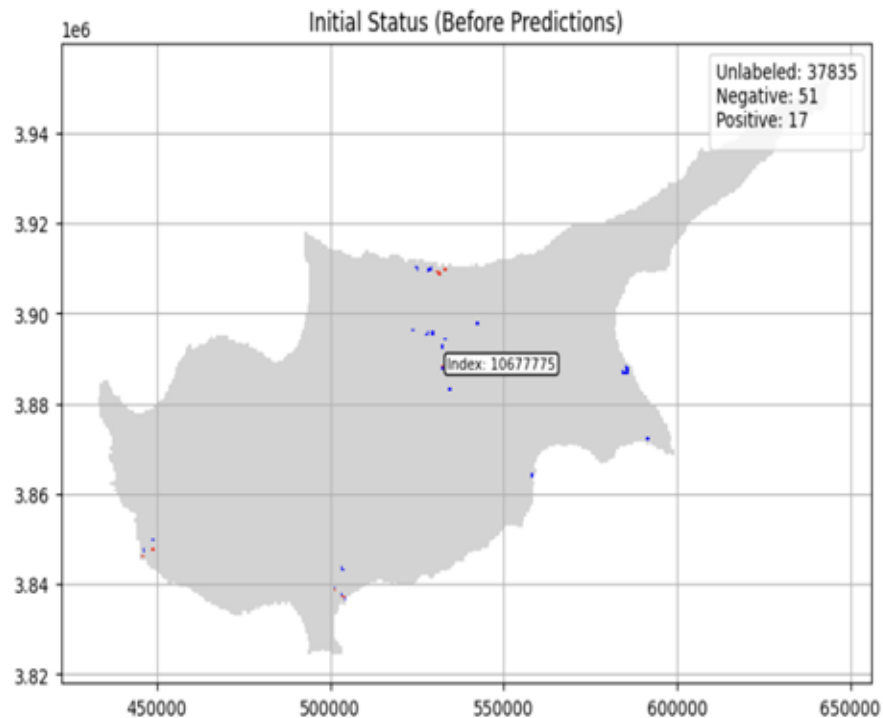


500x500 m cells -> Total 37.9k cells

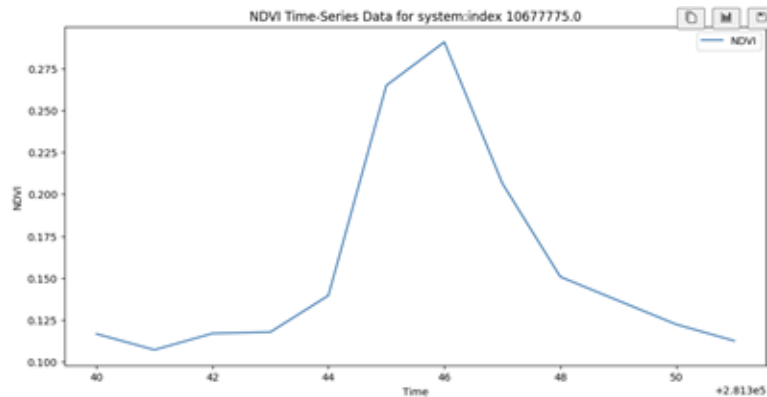
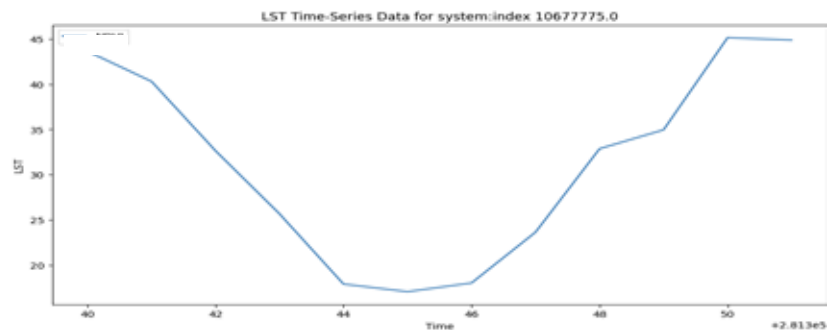
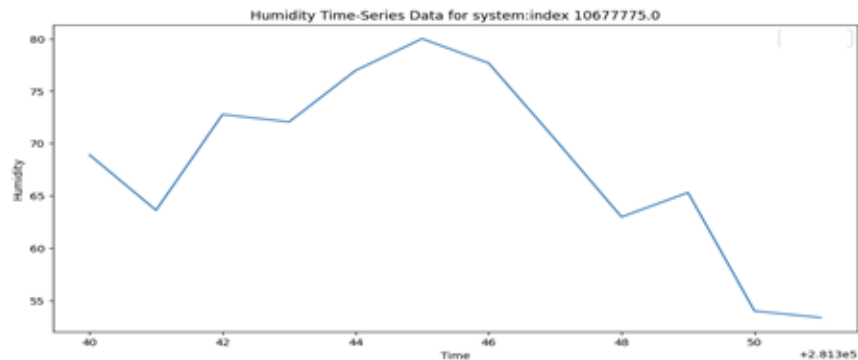
Cell	Time	Elevation	Humidity	LST	NDVI	Nitrogen	OCD	OCS	POP	Precip	SMOD	SOC	WindSpeed	WorldCover	pH
8667760	2023_08	0.422567	82.5459	-	-0.0049	-	-	-	0	0.0635	11	-	4.3753	80	-
	2023_09	0.422567	75.1788	-	-0.0051	-	-	-	0	0.1867	11	-	3.3085	80	-
	2023_10	0.422567	72.5021	-	0.0023	-	-	-	0	1.5087	11	-	3.0965	80	-
	2023_11	0.422567	67.4804	-	0.0007	-	-	-	0	2.0697	11	-	4.476	80	-
	2023_12	0.422567	70.3251	-	-0.0003	-	-	-	0	3.9323	11	-	4.156	80	-
	2024_1	0.422567	66.4759	-	0.0026	-	-	-	0	4.8738	11	-	4.73975	80	-
	2024_2	0.422567	64.51762	-	0.00388	-	-	-	0	6.05358	11	-	5.13195	80	-
	2024_3	0.422567	62.55934	-	0.00516	-	-	-	0	7.23336	11	-	5.52415	80	-
	2024_4	0.422567	60.60106	-	0.00644	-	-	-	0	8.41314	11	-	5.91635	80	-
	2024_5	0.422567	58.64278	-	0.00772	-	-	-	0	9.59292	11	-	6.30855	80	-
	2024_6	0.422567	56.6845	-	0.009	-	-	-	0	10.7727	11	-	6.70075	80	-
	2024_7	0.422567	54.72622	-	0.01028	-	-	-	0	11.95248	11	-	7.09295	80	-

For each cell: 5 time series features and 9 static features.

Pipeline: Prediction Results



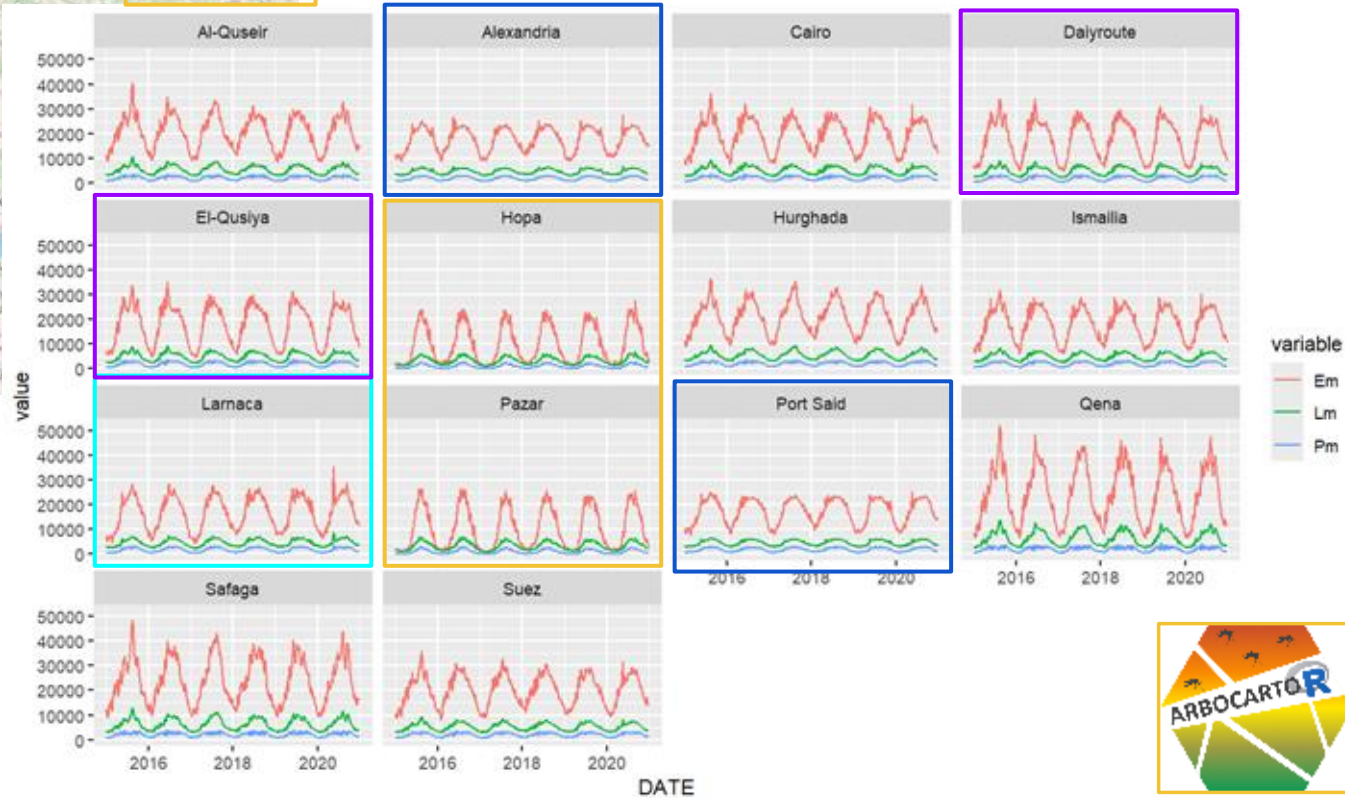
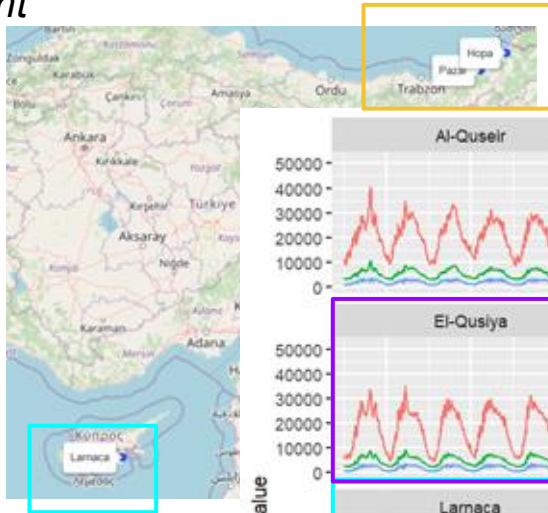
Pipeline: Interpretation of the Results



arbocartoR

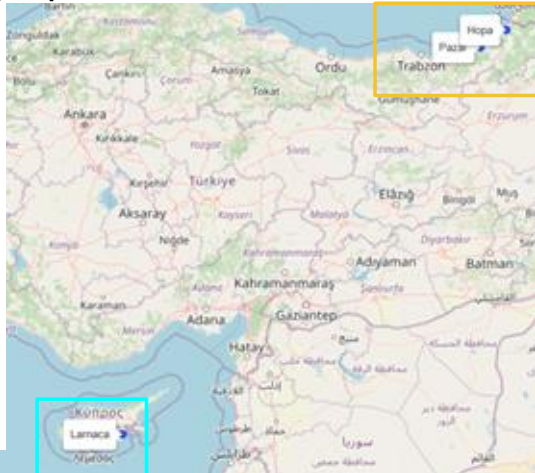
*missing calibration
run on single point*

Eggs (Em)
Larvae (Lm)
Pupae (Pm)

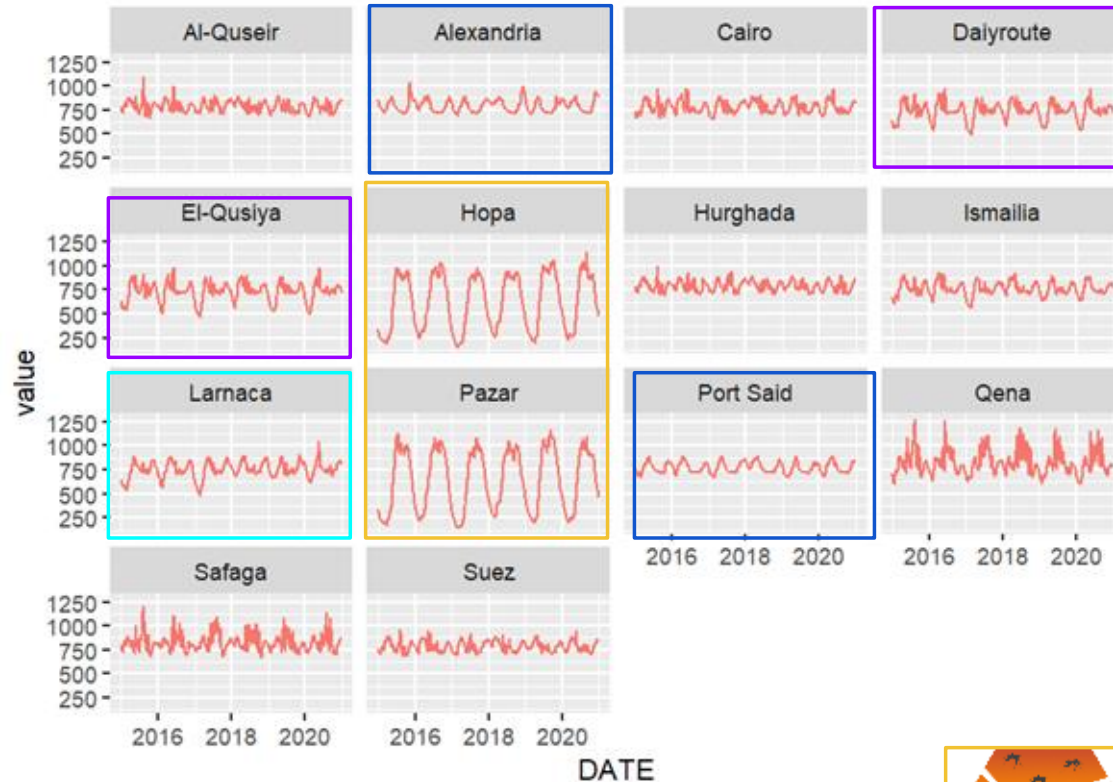


arbocartoR

*missing calibration
run on single point*

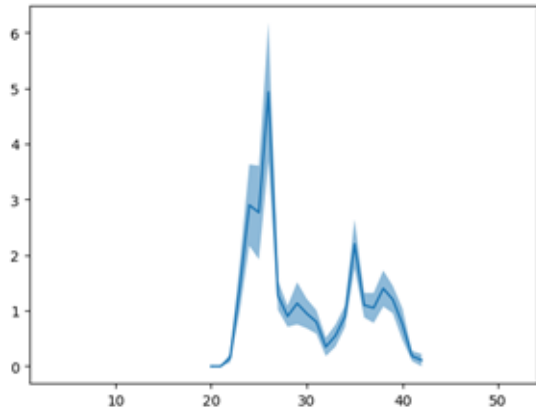


Adults

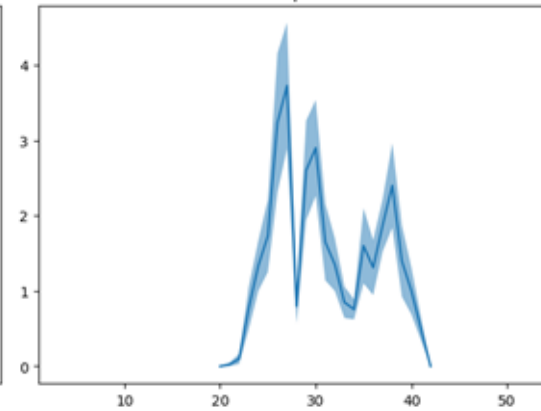


Field tests

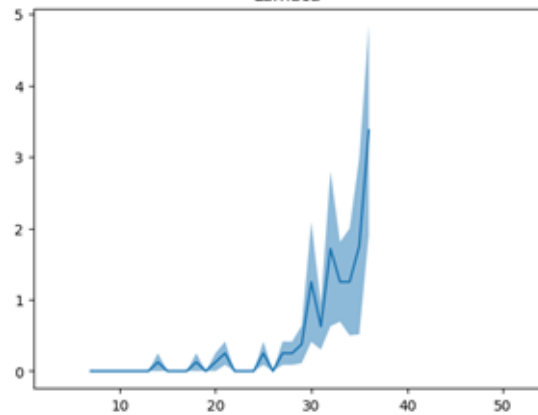
Pazar

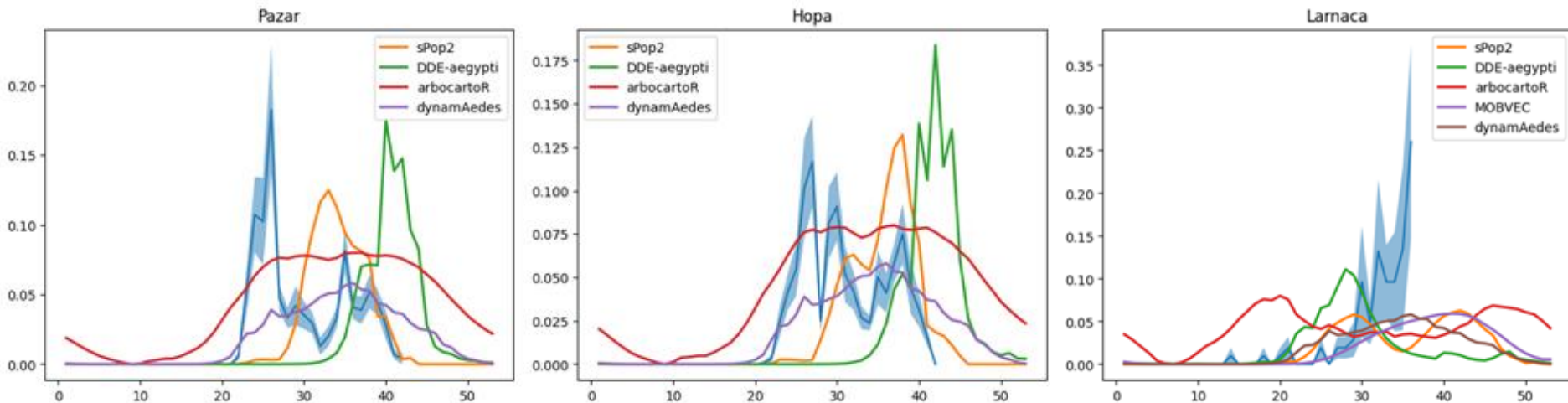


Hopa



Larnaca





Proposed points for qualitative assessment:

- Number of peaks
- Peak location
- Season length
- Season start/end